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IT214-006

Happy learning

This diagram shows a school structure. It is designed to show how the school operates and has all the information from the students, school, teachers and parents.



Conceptual Design:

For this diagram the main entities we are creating are the STUDENT, TEACHER, CLASS, PARENT. For our first relationship we have the TEACHER_AT to CLASS_AT Which is many to many (M:M) because one teacher can be in many classes and a class can have many teachers and this relationship is held by a bridge called ASSIGNMENT_AT, which holds FK: CLASS_AT_CODE_AT and TEACHER_AT_ID_AT. Next one is the PARENT_AT to STUDENT_AT which is a many to many (M:M) relationship because students can have more than one parent and parents can have more than one kid and this relationship is held by a bridge called RELATIONSHIP, which is holding FK: STUDENT_AT_ID_AT and PARENT_AT_ID_AT. We also have the STUDENT_AT to CLASS_AT which is many to one (M:1) because a class can have many kids and kids can only be in one class time. We also have three new entities LUNCH_ORDER_AT, LUNCH_ITEM_AT and ORDER_ITEM_AT. LUNCH_ORDER_AT has a (1:M) relationship to the STUDENT_AT with a FK: STUDENT_AT_ID_AT. Last relation is between LUNCH_ORDER_AT and LUNCH_ITEM_AT that is M:M which has a bridge called ORDER_ITEM_AT.

Entities Analysis:

- STUDNET_AT

- This entity shows the students information table.
- For this entity since there is no PK given we will use a surrogate key ID_AT and the foreign key is CLASS_AT_CODE_AT which comes from CLASS_AT ID CODE_AT.

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
ID_AT	INT	Simple	Single	Required	Unique
FIRST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
MIDDLE_NAME_AT	VARCHAR(15)	Simple	Single	Optional	Duplicated
LAST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
NICK_NAME_AT	VARCHAR(10)	Simple	Single	Required	Duplicated
DATE_OF_BIRTH_AT	DATE	Simple	Single	Required	Duplicated
ADDRESS_AT	VARCHAR(55)	Composite	Single	Required	Duplicated
CITY_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
ZIPCODE_AT	DECIMAL(0,5)	Simple	Single	Required	Duplicated
STREET_AT	VARCHAR(35)	Simple	Single	Required	Duplicated
CLASS_AT_CODE_AT	VARCHAR(2)	Simple	Single	Required	Duplicated

- PARENT_AT

- This entity shows the parent/guardian information table.
- For this entity since there is no PK given, we will use a surrogate key ID_AT and there is no foreign key

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
ID_AT	INT	Simple	Single	Required	Unique
FIRST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
LAST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
MIDDLE_NAME_AT	VARCHAR(15)	Simple	Single	Optional	Duplicated
HOME_PHONE_AT	CHAR(13)	Simple	Single	Optional	Duplicated
MOBILE_PHONE_AT	CHAR(13)	Simple	Single	Required	Duplicated
WORK_PHONE_AT	CHAR(13)	Simple	Single	Optional	Duplicated
EMAIL_AT	VARCHAR(30)	Simple	Single	Optional	Duplicated
RELATION_AT	VARCHAR(20)	Simple	Single	Required	Duplicated

- **CLASS_AT**

- This entity shows the class information table.
- For this entity the primary key is the CODE_AT and there is no foreign key

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
CLASS_CODE_AT	VARCHAR(2)	Composite	Single	Required	Unique
AGE_LEVEL_AT	VARCHAR(7)	Simple	Single	Required	Duplicated
COLOR_AT	VARCHAR(5)	Simple	Single	Required	Duplicated
ROOM_NUMBER_AT	INT	Simple	Single	Required	Duplicated
PHONE_NUMBER_AT	CHAR(13)	Simple	Single	Required	Duplicated
DIRECTION_AT	VARCHAR(160)	Simple	Single	Optional	Duplicated
CAPACITY_AT	INT	Simple	Single	Required	Duplicated

- **TEACHER_AT**

- This entity shows the teacher information table.
- For this entity since there is no PK given, we will use a surrogate key ID_AT and there is no foreign key

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
ID_AT	INT	Simple	Single	Required	Unique
FIRST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
LAST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
MIDDLE_NAME_AT	VARCHAR(15)	Simple	Single	Optional	Duplicated
HOME_PHONE_AT	CHAR(13)	Simple	Single	Required	Duplicated
WORK_EMAIL_AT	VARCHAR(30)	Simple	Single	Required	Duplicated
PERSONAL_EMAIL_AT	VARCHAR(30)	Simple	Single	Optional	Duplicated
COLLEGE_AT	VARCHAR(50)	Simple	Single	Optional	Duplicated
HIGHEST_DEGREE_AT	VARCHAR(5)	Simple	Single	Required	Duplicated
DEGREE_AREA_AT	VARCHAR(25)	Simple	Single	Required	Duplicated

- **RELATIONSHIP_AT**

- This entity shows the relationship between Student & Parent. Since the relationship between Student and Parent is M:M we create a bridge called **RELATIONSHIP** to connect them.

Attribute Name	Domain & Size
STUDENT_AT_ID_AT	INT
PARENT_AT_ID_AT	INT

- **ASSIGNMENT_AT**

- This entity shows the relationship between Class & Teacher. Since the relationship between Class and Teacher is M:M we create a bridge called **ASSIGNMENT** to connect them.

Attribute Name	Domain & Size
CLASS_AT_CODE_AT	INT
TEACHER_AT_ID_AT	INT

- **LUNCH_ORDER_AT**

- This entity shows the Order information table.
- For this entity since there is no PK given, we will use a surrogate key ID_AT and there is a FK: STUDENT_AT_ID_AT references STUDENT_AT(ID_AT)

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
ID_AT	INT	Simple	Single	Required	Unique
QUANTITY_AT	INT	Simple	Single	Required	Duplicated
DATE_AT	DATE	Simple	Single	Required	Duplicated
STUDENT_AT_ID_AT	INT	Simple	Single	Required	Duplicated

- **LUNCH_ITEM_AT**

- This entity shows the item information table.
- we will use a primary key ID_AT, no foreign key

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
CODE_AT	INT	Simple	Single	Required	Unique
ITEM_NAME_AT	VARCHAR(26)	Simple	Single	Required	Duplicated
CALORIES_AT	INT	Simple	Single	Required	Duplicated

- **ORDER_ITEM_AT**

- This entity shows the relationship between **LUNCH_ORDER_AT** & **LUNCH_ITEM_AT**. Since the relationship between LUNCH_ORDER and LUNCH_ITEM is M:M we create a bridge called **ORDER_ITEM_AT** to connect them.

Attribute Name	Domain & Size
LUNCH_ORDER_AT_ID_AT	INT
LUNCH_ITEM_AT_CODE_AT	INT

Relationship Analysis:

→ TEACHER_AT to CLASS_AT

- ◆ We use a bridge : **ASSIGNMENT_AT**
 - TEACHER has class ASSIGNMENT (1:M)
 - CLASS has teacher ASSIGNMENT (1:M)
- ◆ M:M because one teacher can be in many classes and one class can have many teachers.
- ◆ The strength is weak because both entities exist independently, not relying on each other.
- ◆ TEACHER(total) because teachers get assigned to a class and CLASS (total) because all classes have teachers.
- ◆ No special cardinality
- ◆ FK: The ASSIGNMENT table holding the CLASS_AT_CODE_AT reference to the class entity and TEACHER_AT_ID_AT reference to the teacher entity

→ PARENT_AT to STUDENT_AT

- ◆ We use a bridge: **RELATIONSHIP_AT**
 - PARENT has student RELATIONSHIP (1:M)
 - STUDENT has parent RELATIONSHIP (1:M)
- ◆ M:M because students can have more than one parent and parents can have more than one kid
- ◆ The strength is strong because each student is connected to a parent or a caretaker
- ◆ They are both total because they both need each other
- ◆ No special cardinality

- ◆ FK: The RELATIONSHIP table holding the STUDENT_AT_ID_AT reference to the student entity and PARENT_AT_ID_AT reference to the parent entity

→ STUDENT_AT to CLASS_AT

- ◆ STUDENT enrolls in CLASS
- ◆ One CLASS contains many STUDENTS, and one STUDENT is enrolled in one CLASS at a time. No bridge
- ◆ The straight is weak because both entities exist independently, not relying on each other.
- ◆ CLASS (partial) because all the classes might not have kids sometimes, and STUDENTS (total) because a student needs to be always enrolled in a class.
- ◆ No special cardinality
- ◆ FK: CLASS_AT_CODE_AT reference to the CLASS entity

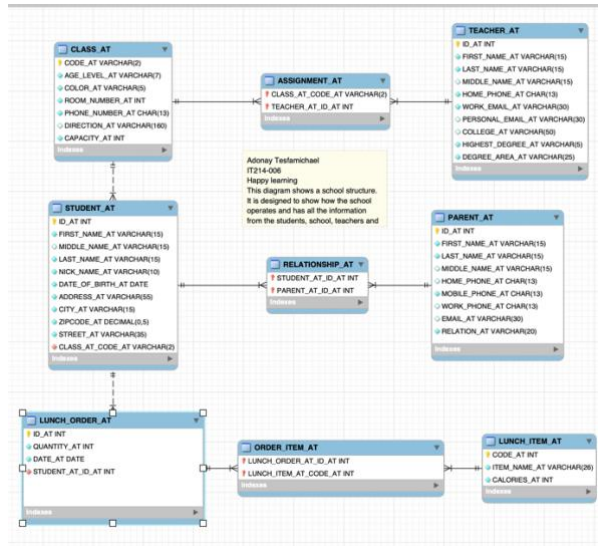
→ STUDENT_AT & LUNCH_ORDER_AT

- ◆ STUDENT_AT orders LUNCH_ORDER_AT
- ◆ 1:M Each student can order one lunch every day, and lunch can be ordered many times every day for many days
- ◆ The straight is weak, because a student might not order lunch.
- ◆ Both is partial because there might not be an order someday to have a menu
- ◆ FK: STUDENT_AT_ID_AT references STUDENT_AT(ID_AT)

→ LUNCH_ORDER_AT & LUNCH_ITEM_AT

- ◆ We use a bridge: ORDER_ITEM_AT
 - LUNCH_ORDER_AT has lunch item ORDER_ITEM_AT(1:M)
 - LUNCH_ITEM_AT has lunch order ORDER_ITEM_AT(1:M)
- ◆ M:M because one order may have up to 4 items and each item will be shows on orders
- ◆ The straight is strong because an order can't be placed without items
- ◆ They are both total because they both need each other
- ◆ FK: The ORDER_ITEM_AT table holding the LUNCH_ORDER_AT_ID_AT reference to the LUNCH_ORDER_AT entity and LUNCH_ITEM_AT_CODE_AT reference to the LUNCH_ITEM_AT entity

Entity Relationship Diagram:



TEACHER_AT

TEACHER_AT - Table					
Name: TEACHER_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
FIRST_NAME_AT	VARCHAR(15)	☐	☒	☐	
LAST_NAME_AT	VARCHAR(15)	☐	☒	☐	
MIDDLE_NAME_AT	VARCHAR(15)	☐	☐	☐	
HOME_PHONE_AT	CHAR(13)	☐	☒	☐	
WORK_EMAIL_AT	VARCHAR(30)	☐	☒	☐	
PERSONAL_EMAIL_AT	VARCHAR(30)	☐	☐	☐	
COLLEGE_AT	VARCHAR(50)	☐	☐	☐	
HIGHEST_DEGREE_AT	VARCHAR(5)	☐	☒	☐	
DEGREE_AREA_AT	VARCHAR(25)	☐	☒	☐	

STUDENT_AT

STUDENT_AT - Table					
Name: STUDENT_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
FIRST_NAME_AT	VARCHAR(15)	☐	☒	☐	
MIDDLE_NAME_AT	VARCHAR(15)	☐	☐	☐	
LAST_NAME_AT	VARCHAR(15)	☐	☒	☐	
NICK_NAME_AT	VARCHAR(10)	☐	☒	☐	
DATE_OF_BIRTH_AT	DATE	☐	☒	☐	
ADDRESS_AT	VARCHAR(55)	☐	☒	☐	
CITY_AT	VARCHAR(15)	☐	☒	☐	
ZIPCODE_AT	DECIMAL(0,5)	☐	☒	☐	
STREET_AT	VARCHAR(35)	☐	☒	☐	
CLASS_AT_CODE_AT	VARCHAR(2)	☒	☒	☐	

PARENT_AT

PARENT_AT - Table					
Name: PARENT_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
FIRST_NAME_AT	VARCHAR(15)	☐	☒	☐	
LAST_NAME_AT	VARCHAR(15)	☐	☒	☐	
MIDDLE_NAME_AT	VARCHAR(15)	☐	☐	☐	
HOME_PHONE_AT	CHAR(13)	☐	☐	☐	
MOBILE_PHONE_AT	CHAR(13)	☐	☒	☐	
WORK_PHONE_AT	CHAR(13)	☐	☐	☐	
EMAIL_AT	VARCHAR(30)	☐	☐	☐	
RELATION_AT	VARCHAR(20)	☐	☒	☐	

CLASS_AT

Name: CLASS_AT					
Column	Datatype	PK	NN	UQ	
CODE_AT	VARCHAR(2)	☒	☒	☒	
AGE_LEVEL_AT	VARCHAR(7)	☐	☒	☐	
COLOR_AT	VARCHAR(5)	☐	☒	☐	
ROOM_NUMBER_AT	INT	☐	☒	☐	
PHONE_NUMBER_AT	CHAR(13)	☐	☒	☐	
DIRECTION_AT	VARCHAR(160)	☐	☐	☐	
CAPACITY_AT	INT	☐	☒	☐	

BRIDGE: RELATIONSHIP_AT

- (STUDENT & PARENT)

RELATIONSHIP_AT - Table					
Name: RELATIONSHIP_AT					
Column	Datatype	PK	NN	UQ	
STUDENT_AT_ID_AT	INT	☒	☒	☐	
PARENT_AT_ID_AT	INT	☒	☒	☐	

BRIDGE: ASSIGNMENT_AT

- (CLASS & TEACHER)

ASSIGNMENT_AT - Table				
Name: ASSIGNMENT_AT				
Column	Datatype	PK	NN	
CLASS_AT_CODE_AT	VARCHAR(2)	☒	☒	
TEACHER_AT_ID_AT	INT	☒	☒	

LUNCH_ORDER_AT

Name: LUNCH_ORDER_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
QUANTITY_AT	INT	☐	☒	☐	
DATE_AT	DATE	☐	☒	☐	
STUDENT_AT_ID_AT	INT	☐	☒	☐	

BRIDGE: ORDER_ITEM_AT

- (LUNCH_ORDER & LUNCH_ITEM)

Name: ORDER_ITEM_AT					
Column	Datatype	PK	NN	UQ	
LUNCH_ORDER_AT_ID_AT	INT	☒	☒	☐	
LUNCH_ITEM_AT_CODE_AT	INT	☒	☒	☐	

LUNCH_ITEM_AT

Name: LUNCH_ITEM_AT					
Column	Datatype	PK	NN	UQ	
CODE_AT	INT	☒	☒	☒	
ITEM_NAME_AT	VARCHAR(26)	☐	☒	☐	
CALORIES_AT	INT	☐	☒	☐	

Task 2 - Create database

```
1 -- Task 2 - Create database ADONAY TESFAMICHAEL
2 DROP DATABASE IF EXISTS HAPPY_LEARNING_AT;
3 CREATE DATABASE IF NOT EXISTS HAPPY_LEARNING_AT;
4 USE HAPPY_LEARNING_AT;
5 SHOW DATABASES;
```

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Result Grid Filter Rows: Search Export

Database
FITNESS_COMPANY_AT
HAPPY_LEARNING_AT
HAPPY_TIME
HRL_AT
information_schema
mysql
new_db
performance_schema
S1_A1_Michael_Adonay
StrucDB
Student_Registration_AT
sys

Result 1 Read On

Action Output	Time	Action	Response	Duration / Fetch Time
1	20:51:23	DROP DATABASE IF EXISTS HAPPY_LEARNING_AT	0 row(s) affected	0.108 sec
2	20:51:27	CREATE DATABASE IF NOT EXISTS HAPPY_LEARNING_AT	1 row(s) affected	0.0082 sec
3	20:51:30	USE HAPPY_LEARNING_AT	0 row(s) affected	0.0014 sec
4	20:52:18	SHOW DATABASES	12 row(s) returned	0.0041 sec / 0.00002

Task 3 - Create Table TEACHER

```
8 * CREATE TABLE TEACHER_AT(  
9   ID_AT INT PRIMARY KEY NOT NULL AUTO_INCREMENT,  
10  FIRST_NAME_AT VARCHAR(15) NOT NULL,  
11  MIDDLE_NAME_AT VARCHAR(15),  
12  LAST_NAME_AT VARCHAR(15) NOT NULL,  
13  HOME_PHONE_AT CHAR(13) NOT NULL,  
14  WORK_EMAIL_AT VARCHAR(30) NOT NULL,  
15  PERSONAL_EMAIL_AT VARCHAR(30),  
16  COLLEGE_AT VARCHAR(50),  
17  HIGHEST_DEGREE_AT VARCHAR(5) NOT NULL,  
18  DEGREE_AREA_AT VARCHAR(25) NOT NULL  
19 );  
20 * DESCRIBE TEACHER_AT;
```

Field	Type	Null	Key	Default	Extra
ID_AT	int	NO	PRI		auto_increment
FIRST_NAME_AT	varchar(15)	NO			
MIDDLE_NAME_AT	varchar(15)	YES			
LAST_NAME_AT	varchar(15)	NO			
HOME_PHONE_AT	char(13)	NO			
WORK_EMAIL_AT	varchar(30)	NO			
PERSONAL_EMAIL_AT	varchar(30)	YES			
COLLEGE_AT	varchar(50)	YES			
HIGHEST_DEGREE_AT	varchar(5)	NO			
DEGREE_AREA_AT	varchar(25)	NO			

Result 154

Time	Action	Response	Duration / Fetch Time
15:02:27	CREATE TABLE TEACHER_AT (ID_AT INT PRIMARY KEY NOT NULL A...	0 row(s) affected	0.027 sec
15:02:31	DESCRIBE TEACHER_AT	10 row(s) returned	0.024 sec / 0.000027...

CLASS

```
22 * CREATE TABLE CLASS_AT(  
23  CODE_AT VARCHAR(2) PRIMARY KEY NOT NULL,  
24  AGE_LEVEL_AT VARCHAR(7) NOT NULL,  
25  COLOR_AT VARCHAR(5) NOT NULL,  
26  ROOM_NUMBER_AT INT NOT NULL,  
27  PHONE_NUMBER_AT CHAR(13) NOT NULL,  
28  DIRECTION_AT VARCHAR(160),  
29  CAPACITY_AT INT NOT NULL  
30 );  
31 * DESCRIBE CLASS_AT;
```

Field	Type	Null	Key	Default	Extra
CODE_AT	varchar(2)	NO	PRI		
AGE_LEVEL_AT	varchar(7)	NO			
COLOR_AT	varchar(5)	NO			
ROOM_NUMBER_AT	int	NO			
PHONE_NUMBER_AT	char(13)	NO			
DIRECTION_AT	varchar(160)	YES			
CAPACITY_AT	int	NO			

Result 155

Time	Action	Response	Duration / Fetch Time
15:02:27	CREATE TABLE TEACHER_AT (ID_AT INT PRIMARY KEY NOT NULL A...	0 row(s) affected	0.027 sec
15:02:31	DESCRIBE TEACHER_AT	10 row(s) returned	0.024 sec / 0.000027...
15:03:05	CREATE TABLE CLASS_AT (CODE_AT VARCHAR(2) PRIMARY KEY N...	0 row(s) affected	0.072 sec
15:04:03	DESCRIBE CLASS_AT	7 row(s) returned	0.040 sec / 0.0013 sec

ASSIGNMENT

```
33 * CREATE TABLE ASSIGNMENT_AT(  
34  CLASS_AT_CODE_AT VARCHAR(2) NOT NULL,  
35  TEACHER_AT_ID_AT INT NOT NULL,  
36  FOREIGN KEY (CLASS_AT_CODE_AT) REFERENCES CLASS_AT(CODE_AT),  
37  FOREIGN KEY (TEACHER_AT_ID_AT) REFERENCES TEACHER_AT(ID_AT)  
38 );  
39 * DESCRIBE ASSIGNMENT_AT;
```

Field	Type	Null	Key	Default	Extra
CLASS_AT_CODE_AT	varchar(2)	NO	MUL		
TEACHER_AT_ID_AT	int	NO	MUL		

Result 156

Time	Action	Response	Duration / Fetch Time
15:02:27	CREATE TABLE TEACHER_AT (ID_AT INT PRIMARY KEY NOT NULL A...	0 row(s) affected	0.027 sec
15:02:31	DESCRIBE TEACHER_AT	10 row(s) returned	0.024 sec / 0.000027...
15:03:05	CREATE TABLE CLASS_AT (CODE_AT VARCHAR(2) PRIMARY KEY N...	0 row(s) affected	0.072 sec
15:04:03	DESCRIBE CLASS_AT	7 row(s) returned	0.040 sec / 0.0013 sec
15:04:58	CREATE TABLE ASSIGNMENT_AT (CLASS_AT_CODE_AT VARCHAR...	0 row(s) affected	0.030 sec
15:05:01	DESCRIBE ASSIGNMENT_AT	2 row(s) returned	0.0076 sec / 0.00004...

STUDENT

```
41 * CREATE TABLE STUDENT_AT(
42     ID_AT INT PRIMARY KEY NOT NULL AUTO_INCREMENT,
43     FIRST_NAME_AT VARCHAR(15) NOT NULL,
44     MIDDLE_NAME_AT VARCHAR(15),
45     LAST_NAME_AT VARCHAR(15) NOT NULL,
46     NICK_NAME_AT VARCHAR(10),
47     DATE_OF_BIRTH_AT DATE NOT NULL,
48     ADDRESS_AT VARCHAR(55) NOT NULL,
49     CITY_AT VARCHAR(15) NOT NULL,
50     ZIPCODE_AT DECIMAL(5,0) NOT NULL,
51     STREET_AT VARCHAR(25) NOT NULL,
52     CLASS_AT_CODE_AT VARCHAR(1) NOT NULL,
53     FOREIGN KEY (CLASS_AT_CODE_AT) REFERENCES CLASS_AT(CODE_AT)
54 );
55 * DESCRIBE STUDENT_AT;
```

Field	Type	Null	Key	Default	Extra
ID_AT	int	NO	PK		auto_increment
FIRST_NAME_AT	varchar(15)	NO			
MIDDLE_NAME_AT	varchar(15)	YES			
LAST_NAME_AT	varchar(15)	NO			
NICK_NAME_AT	varchar(10)	YES			
DATE_OF_BIRTH_AT	date	NO			
ADDRESS_AT	varchar(55)	NO			
CITY_AT	varchar(15)	NO			
ZIPCODE_AT	decimal(5,0)	NO			
STREET_AT	varchar(25)	NO			
CLASS_AT_CODE_AT	varchar(2)	NO	MUL		

Time	Action	Response	Duration / Fetch Time
15-08-31	CREATE TABLE STUDENT_AT(ID_AT INT PRIMARY KEY NOT NULL A...	0 row(s) affected	0.029 sec
15-08-34	DESCRIBE STUDENT_AT	11 row(s) returned	0.0048 sec / 0.0000...

PARENT

```
57 * CREATE TABLE PARENT_AT(
58     ID_AT INT PRIMARY KEY NOT NULL AUTO_INCREMENT,
59     FIRST_NAME_AT VARCHAR(15) NOT NULL,
60     MIDDLE_NAME_AT VARCHAR(15),
61     LAST_NAME_AT VARCHAR(15) NOT NULL,
62     HOME_PHONE_AT CHAR(13),
63     MOBILE_PHONE_AT CHAR(13) NOT NULL,
64     WORK_PHONE_AT CHAR(13),
65     EMAIL_AT VARCHAR(30),
66     RELATION VARCHAR(20) NOT NULL
67 );
68 * DESCRIBE PARENT_AT;
```

Field	Type	Null	Key	Default	Extra
ID_AT	int	NO	PK		auto_increment
FIRST_NAME_AT	varchar(15)	NO			
MIDDLE_NAME_AT	varchar(15)	YES			
LAST_NAME_AT	varchar(15)	NO			
HOME_PHONE_AT	char(13)	YES			
MOBILE_PHONE_AT	char(13)	NO			
WORK_PHONE_AT	char(13)	YES			
EMAIL_AT	varchar(30)	YES			
RELATION	varchar(20)	NO			

Time	Action	Response	Duration / Fetch Time
15-08-31	CREATE TABLE STUDENT_AT(ID_AT INT PRIMARY KEY NOT NULL A...	0 row(s) affected	0.029 sec
15-08-34	DESCRIBE STUDENT_AT	11 row(s) returned	0.0048 sec / 0.0000...
15-07-36	CREATE TABLE PARENT_AT(ID_AT INT PRIMARY KEY NOT NULL AU...	0 row(s) affected	0.032 sec
15-07-39	DESCRIBE PARENT_AT	9 row(s) returned	0.0081 sec / 0.00005...

RELATIONSHIP

```
70 * CREATE TABLE RELATIONSHIP_AT(
71     STUDENT_AT_ID_AT INT NOT NULL,
72     PARENT_AT_ID_AT INT NOT NULL,
73     FOREIGN KEY (STUDENT_AT_ID_AT) REFERENCES STUDENT_AT(ID_AT),
74     FOREIGN KEY (PARENT_AT_ID_AT) REFERENCES PARENT_AT(ID_AT)
75 );
76 * DESCRIBE RELATIONSHIP_AT;
```

Field	Type	Null	Key	Default	Extra
STUDENT_AT_ID_AT	int	NO	MUL		
PARENT_AT_ID_AT	int	NO	MUL		

Time	Action	Response	Duration / Fetch Time
15-08-31	CREATE TABLE STUDENT_AT(ID_AT INT PRIMARY KEY NOT NULL A...	0 row(s) affected	0.029 sec
15-08-34	DESCRIBE STUDENT_AT	11 row(s) returned	0.0048 sec / 0.0000...
15-07-36	CREATE TABLE PARENT_AT(ID_AT INT PRIMARY KEY NOT NULL AU...	0 row(s) affected	0.032 sec
15-07-39	DESCRIBE PARENT_AT	9 row(s) returned	0.0081 sec / 0.00005...
15-08-11	CREATE TABLE RELATIONSHIP_AT(STUDENT_AT_ID_AT INT NOT N...	0 row(s) affected	0.033 sec
15-08-14	DESCRIBE RELATIONSHIP_AT	2 row(s) returned	0.0071 sec / 0.00003...

LUNCH_ORDER_AT

```
78 * CREATE TABLE LUNCH_ORDER_AT(  
79   ID_AT INT PRIMARY KEY NOT NULL AUTO_INCREMENT,  
80   QUANTITY_AT INT NOT NULL,  
81   DATE_AT DATE NOT NULL,  
82   STUDENT_AT_ID_AT INT NOT NULL,  
83   FOREIGN KEY (STUDENT_AT_ID_AT) REFERENCES STUDENT_AT(ID_AT)  
84 );  
85 * DESCRIBE LUNCH_ORDER_AT;  
86
```

Field	Type	Null	Key	Default	Extra
ID_AT	int	NO	PK		auto_increment
QUANTITY_AT	int	NO			
DATE_AT	date	NO			
STUDENT_AT_ID_AT	int	NO	MUL		

Result 160

Time	Action	Response	Duration / Fetch Time
15:09:03	CREATE TABLE LUNCH_ORDER_AT(ID_AT INT PRIMARY KEY NOT N...	0 row(s) affected	0.026 sec
15:09:04	DESCRIBE LUNCH_ORDER_AT	4 row(s) returned	0.0046 sec / 0.0000...

LUNCH_ITEM_AT

```
87 * CREATE TABLE LUNCH_ITEM_AT(  
88   CODE_AT INT PRIMARY KEY NOT NULL AUTO_INCREMENT,  
89   ITEM_NAME_AT VARCHAR(26) NOT NULL,  
90   CALORIES INT NOT NULL  
91 );  
92 * DESCRIBE LUNCH_ITEM_AT;  
93
```

Field	Type	Null	Key	Default	Extra
CODE_AT	int	NO	PK		auto_increment
ITEM_NAME_AT	varchar(26)	NO			
CALORIES	int	NO			

Result 161

Time	Action	Response	Duration / Fetch Time
15:09:03	CREATE TABLE LUNCH_ORDER_AT(ID_AT INT PRIMARY KEY NOT N...	0 row(s) affected	0.026 sec
15:09:04	DESCRIBE LUNCH_ORDER_AT	4 row(s) returned	0.0046 sec / 0.0000...
15:11:39	CREATE TABLE LUNCH_ITEM_AT(CODE_AT INT PRIMARY KEY NOT ...	0 row(s) affected	0.064 sec
15:11:41	DESCRIBE LUNCH_ITEM_AT	3 row(s) returned	0.033 sec / 0.00038...

ORDER_ITEM_AT

```
94 * CREATE TABLE ORDER_ITEM_AT(  
95   LUNCH_ORDER_AT_ID_AT INT NOT NULL,  
96   LUNCH_ITEM_AT_CODE_AT INT NOT NULL,  
97   FOREIGN KEY (LUNCH_ORDER_AT_ID_AT) REFERENCES LUNCH_ORDER_AT(ID_AT),  
98   FOREIGN KEY (LUNCH_ITEM_AT_CODE_AT) REFERENCES LUNCH_ITEM_AT(CODE_AT)  
99 );  
100 * DESCRIBE ORDER_ITEM_AT;  
101
```

Field	Type	Null	Key	Default	Extra
LUNCH_ORDER_AT_ID_AT	int	NO	MUL		
LUNCH_ITEM_AT_CODE_AT	int	NO	MUL		

Result 162

Time	Action	Response	Duration / Fetch Time
15:09:03	CREATE TABLE LUNCH_ORDER_AT(ID_AT INT PRIMARY KEY NOT N...	0 row(s) affected	0.026 sec
15:09:04	DESCRIBE LUNCH_ORDER_AT	4 row(s) returned	0.0046 sec / 0.0000...
15:11:39	CREATE TABLE LUNCH_ITEM_AT(CODE_AT INT PRIMARY KEY NOT ...	0 row(s) affected	0.064 sec
15:11:41	DESCRIBE LUNCH_ITEM_AT	3 row(s) returned	0.033 sec / 0.00038...
15:12:28	CREATE TABLE ORDER_ITEM_AT(LUNCH_ORDER_AT_ID_AT INT N...	0 row(s) affected	0.020 sec
15:12:30	DESCRIBE ORDER_ITEM_AT	2 row(s) returned	0.0084 sec / 0.0000...

Task 4 - Add data

TEACHER

```
102 -- Task 4 - Add data ADONAY TESFAMICHAEL  
103 * INSERT INTO TEACHER_AT VALUES  
104 (1,'Katheryn',NULL,'Tsfamichael', '(123) 4567890', 'kat@work.email','kat@personal.email','George Mason University', 'PhD','Computer Science'),  
105 (2,'John','A','Smith','(234) 5678901', 'john.smith@work.edu', 'jsmith@gmail.com','University of Virginia','BS', 'Early Education'),  
106 (3,'Emily', NULL,'Clark','(345) 6789012', 'eclark@work.edu', 'emily.clark@gmail.com','George Mason University', 'MS', 'Mathematics');  
107 * SELECT * FROM TEACHER_AT;  
108
```

ID_AT	FIRST_NAME_AT	MIDDLE_NAME_AT	LAST_NAME_AT	HOME_PHONE_AT	WORK_EMAIL_AT	PERSONAL_EMAIL_AT	COLLEGE_AT	HIGHEST_DEGREE_AT	DEGREE_AREA_AT
1	Katheryn		Tsfamichael	(123) 4567890	kat@work.email	kat@personal.email	George Mason University	PhD	Computer Science
2	John	A	Smith	(234) 5678901	john.smith@work.edu	jsmith@gmail.com	University of Virginia	BS	Early Education
3	Emily		Clark	(345) 6789012	eclark@work.edu	emily.clark@gmail.com	George Mason University	MS	Mathematics

TEACHER_AT 163

Time	Action	Response	Duration / Fetch Time
15:19:03	INSERT INTO TEACHER_AT VALUES (1,'Katheryn',NULL,'Tsfamichae...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.070 sec
15:19:05	SELECT * FROM TEACHER_AT LIMIT 0, 1000	3 row(s) returned	0.011 sec / 0.0011 sec

CLASS

```
109 • INSERT INTO CLASS_AT VALUES
110 ('G1', 'foxes', 'GF', 101, (777) 7890123, 'North wing, near front door', 20),
111 ('P1', 'monkeys', 'PM', 202, (999) 0123456, 'East wing, 2nd floor', 18),
112 ('R2', 'turtles', 'RT', 103, (111) 1234567, 'West wing, 1st floor', 22);
113 • SELECT * FROM CLASS_AT;
```

CODE_AT	AGE_LEVEL_AT	COLOR_AT	ROOM_NUMBER_AT	PHONE_NUMBER_AT	DIRECTION_AT	CAPACITY_AT
G1	foxes	GF	101	(777) 7890123	North wing, near front door	20
P1	monkeys	PM	202	(999) 0123456	East wing, 2nd floor	18
R2	turtles	RT	103	(111) 1234567	West wing, 1st floor	22

CLASS_AT 164

Time	Action	Response	Duration / Fetch Time
15:19:03	INSERT INTO TEACHER_AT VALUES ('Katheryn', NULL, 'Tesfamichae...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.070 sec
15:19:05	SELECT * FROM TEACHER_AT LIMIT 0, 1000	3 row(s) returned	0.011 sec / 0.0011 sec
15:19:37	INSERT INTO CLASS_AT VALUES ('G1', 'foxes', 'GF', 101, (777) 789012...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.018 sec
15:19:40	SELECT * FROM CLASS_AT LIMIT 0, 1000	3 row(s) returned	0.0013 sec / 0.00005...

ASSIGNMENT

```
115 • INSERT INTO ASSIGNMENT_AT VALUES
116 ('G1', 1),
117 ('P1', 2),
118 ('R2', 3);
119 • SELECT * FROM ASSIGNMENT_AT;
```

CLASS_AT_CODE_AT	TEACHER_AT_ID_AT
G1	1
P1	2
R2	3

ASSIGNMENT_AT 165

Time	Action	Response	Duration / Fetch Time
15:19:03	INSERT INTO TEACHER_AT VALUES ('Katheryn', NULL, 'Tesfamichae...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.070 sec
15:19:05	SELECT * FROM TEACHER_AT LIMIT 0, 1000	3 row(s) returned	0.011 sec / 0.0011 sec
15:19:37	INSERT INTO CLASS_AT VALUES ('G1', 'foxes', 'GF', 101, (777) 789012...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.018 sec
15:19:40	SELECT * FROM CLASS_AT LIMIT 0, 1000	3 row(s) returned	0.0013 sec / 0.00005...
15:20:24	INSERT INTO ASSIGNMENT_AT VALUES ('G1', 1), ('P1', 2), ('R2', 3)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.012 sec
15:20:26	SELECT * FROM ASSIGNMENT_AT LIMIT 0, 1000	3 row(s) returned	0.00069 sec / 0.000...

STUDENT

```
121 • INSERT INTO STUDENT_AT VALUES
122 (1, 'Alice', NULL, 'Johnson', 'Ali', '2015-08-15', '123 Maple St', 'Fairfax', 22030, 'Maple Street', 'G1'),
123 (2, 'Bob', 'Michael', 'Williams', NULL, '2016-05-20', '456 Oak Ave', 'Falls Church', 22048, 'Oak Avenue', 'P1'),
124 (3, 'Carol', 'Ann', 'Davis', 'Cara', '2013-11-10', '789 Pine Rd', 'Arlington', 22201, 'Pine Road', 'R2');
125 • SELECT * FROM STUDENT_AT;
```

ID_AT	FIRST_NAME_AT	MIDDLE_NAME_AT	LAST_NAME_AT	NICK_NAME_AT	DATE_OF_BIRTH_AT	ADDRESS_AT	CITY_AT	ZIPCODE_AT	STREET_AT	CLASS_AT_CODE_AT
1	Alice		Johnson	Ali	2015-08-15	123 Maple St	Fairfax	22030	Maple Street	G1
2	Bob	Michael	Williams		2016-05-20	456 Oak Ave	Falls Church	22048	Oak Avenue	P1
3	Carol	Ann	Davis	Cara	2013-11-10	789 Pine Rd	Arlington	22201	Pine Road	R2

STUDENT_AT 166

Time	Action	Response	Duration / Fetch Time
15:19:03	INSERT INTO TEACHER_AT VALUES ('Katheryn', NULL, 'Tesfamichae...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.070 sec
15:19:05	SELECT * FROM TEACHER_AT LIMIT 0, 1000	3 row(s) returned	0.011 sec / 0.0011 sec
15:19:37	INSERT INTO CLASS_AT VALUES ('G1', 'foxes', 'GF', 101, (777) 789012...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.018 sec
15:19:40	SELECT * FROM CLASS_AT LIMIT 0, 1000	3 row(s) returned	0.0013 sec / 0.00005...
15:20:24	INSERT INTO ASSIGNMENT_AT VALUES ('G1', 1), ('P1', 2), ('R2', 3)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.012 sec
15:20:26	SELECT * FROM ASSIGNMENT_AT LIMIT 0, 1000	3 row(s) returned	0.00069 sec / 0.000...
15:20:58	INSERT INTO STUDENT_AT VALUES (1, 'Alice', NULL, 'Johnson', 'Ali', '2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.018 sec
15:21:02	SELECT * FROM STUDENT_AT LIMIT 0, 1000	3 row(s) returned	0.00095 sec / 0.000...

PARENT

```
127 • INSERT INTO PARENT_AT VALUES
128 (1, 'Jane', NULL, 'Johnson', '(111) 2223333', '(111) 4445555', '(111) 6667777', 'johnson@parent.com', 'Mother'),
129 (2, 'William', 'JR', 'Williams', '(222) 3334444', '(222) 5556666', '(222) 7778888', 'williams@parent.com', 'Father'),
130 (3, 'Susan', NULL, 'Davis', '(333) 4445555', '(333) 5556666', NULL, 'davis@parent.com', 'Mother');
131 • SELECT * FROM PARENT_AT;
```

ID_AT	FIRST_NAME_AT	MIDDLE_NAME_AT	LAST_NAME_AT	HOME_PHONE_AT	MOBILE_PHONE_AT	WORK_PHONE_AT	EMAIL_AT	RELATION
1	Jane		Johnson	(111) 2223333	(111) 4445555	(111) 6667777	johnson@parent.com	Mother
2	William	JR	Williams	(222) 3334444	(222) 5556666	(222) 7778888	williams@parent.com	Father
3	Susan		Davis	(333) 4445555	(333) 5556666		davis@parent.com	Mother

PARENT_AT 167

Time	Action	Response	Duration / Fetch Time
15:21:36	INSERT INTO PARENT_AT VALUES (1, 'Jane', NULL, 'Johnson', '(111)...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.011 sec
15:21:38	SELECT * FROM PARENT_AT LIMIT 0, 1000	3 row(s) returned	0.0012 sec / 0.00003...

RELATIONSHIP

```
133 • INSERT INTO RELATIONSHIP_AT VALUES
134 (1, 1),
135 (2, 2),
136 (3, 3);
137 • SELECT * FROM RELATIONSHIP_AT;
```

STUDENT_AT_ID_AT	PARENT_AT_ID_AT
1	1
2	2
3	3

RELATIONSHIP_AT 168

Time	Action	Response	Duration / Fetch Time
15:21:36	INSERT INTO PARENT_AT VALUES (1, 'Jane', NULL, 'Johnson', '(111)...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.011 sec
15:21:38	SELECT * FROM PARENT_AT LIMIT 0, 1000	3 row(s) returned	0.0012 sec / 0.00003...
15:22:06	INSERT INTO RELATIONSHIP_AT VALUES (1, 1), (2, 2), (3, 3)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.0070 sec
15:22:08	SELECT * FROM RELATIONSHIP_AT LIMIT 0, 1000	3 row(s) returned	0.00094 sec / 0.000...

LUNCH_ITEM_AT

```
INSERT INTO LUNCH_ITEM_AT VALUES
140 (1, 'Pizza ', 200),
141 (2, 'Sandwich', 300),
142 (3, 'Salad', 50),
143 (4, 'Apple Juice', 100);
144 • SELECT * FROM LUNCH_ITEM_AT;
```

CODE_AT	ITEM_NAME_AT	CALORIES
1	Pizza	200
2	Sandwich	300
3	Salad	50
4	Apple Juice	100

LUNCH_ITEM_AT 169

Time	Action	Response	Duration / Fetch Time
15:21:36	INSERT INTO PARENT_AT VALUES (1, 'Jane', NULL, 'Johnson', '(111)...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.011 sec
15:21:38	SELECT * FROM PARENT_AT LIMIT 0, 1000	3 row(s) returned	0.0012 sec / 0.00003...
15:22:06	INSERT INTO RELATIONSHIP_AT VALUES (1, 1), (2, 2), (3, 3)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.0070 sec
15:22:08	SELECT * FROM RELATIONSHIP_AT LIMIT 0, 1000	3 row(s) returned	0.00094 sec / 0.000...
15:22:36	INSERT INTO LUNCH_ITEM_AT VALUES (1, 'Pizza ', 200), (2, 'Sandw...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.014 sec
15:22:38	SELECT * FROM LUNCH_ITEM_AT LIMIT 0, 1000	4 row(s) returned	0.00097 sec / 0.0000...

LUNCH_ORDER_AT

```
146 • INSERT INTO LUNCH_ORDER_AT VALUES
147 (1, 2, '2025-05-01', 1),
148 (2, 2, '2025-05-03', 2),
149 (3, 1, '2025-05-05', 3);
150 • SELECT * FROM LUNCH_ORDER_AT;
```

100% 30:150

Result Grid Filter Rows: Search Edit: Export/Import:

ID_AT	QUANTITY_AT	DATE_AT	STUDENT_AT_ID_AT
1	2	2025-05-01	1
2	2	2025-05-03	2
3	1	2025-05-05	3

LUNCH_ORDER_AT 171 Apply

Action Output

	Time	Action	Response	Duration / Fetch Time
1	15:23:23	INSERT INTO LUNCH_ORDER_AT VALUES (1, 2, '2025-05-01', 1), (2, ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.0083 sec
2	15:23:24	SELECT * FROM LUNCH_ORDER_AT LIMIT 0, 1000	3 row(s) returned	0.00090 sec / 0.000...

ORDER_ITEM_AT

```
INSERT INTO ORDER_ITEM_AT VALUES
153 (1, 1),
154 (1, 4),
155 (2, 2),
156 (2, 4),
157 (3, 3);
158 • SELECT * FROM ORDER_ITEM_AT;
```

100% 29:158

Result Grid Filter Rows: Search Export:

LUNCH_ORDER_AT_ID_AT	LUNCH_ITEM_AT_CODE_AT
1	1
1	4
2	2
2	4
3	3

ORDER_ITEM_AT 172 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
1	15:23:23	INSERT INTO LUNCH_ORDER_AT VALUES (1, 2, '2025-05-01', 1), (2, ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.0083 sec
2	15:23:24	SELECT * FROM LUNCH_ORDER_AT LIMIT 0, 1000	3 row(s) returned	0.00090 sec / 0.000...
3	15:23:56	INSERT INTO ORDER_ITEM_AT VALUES (1, 1), (1, 4), (2, 2), (2, 4), (3, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.0089 sec
4	15:23:58	SELECT * FROM ORDER_ITEM_AT LIMIT 0, 1000	5 row(s) returned	0.00099 sec / 0.000...

Task 5 Query the data

1.

```
160 -- Task 5 - Query the data ADONAY TESFAMICHAEL
161 -- List all students names (in a column named 'Student') in alphabetical order, with the age level (in a column named 'Level'), color and directions of the cl
162 • SELECT CONCAT_WS(' ', STUDENT_AT.FIRST_NAME_AT, STUDENT_AT.MIDDLE_NAME_AT, STUDENT_AT.LAST_NAME_AT) AS Student,
163 CLASS_AT.AGE_LEVEL_AT AS Level,
164 CLASS_AT.COLOR_AT AS Color,
165 CLASS_AT.DIRECTION_AT AS Direction,
166 CONCAT_WS(' ', TEACHER_AT.FIRST_NAME_AT, TEACHER_AT.MIDDLE_NAME_AT, TEACHER_AT.LAST_NAME_AT) AS Teacher
167 FROM STUDENT_AT
168 JOIN CLASS_AT ON STUDENT_AT.CLASS_AT_CODE_AT = CLASS_AT.CODE_AT
169 JOIN ASSIGNMENT_AT ON CLASS_AT.CODE_AT = ASSIGNMENT_AT.CLASS_AT_CODE_AT
170 JOIN TEACHER_AT ON ASSIGNMENT_AT.TEACHER_AT_ID_AT = TEACHER_AT.ID_AT
171 ORDER BY Student ASC;
```

100% 22:171

Result Grid Filter Rows: Search Export:

Student	Level	Color	Direction	Teacher
Alice Johnson	foxes	GF	North wing, near front door	Katheryn Tesfamichael
Bob Michael Williams	monkeys	PM	East wing, 2nd floor	John A Smith
Carol Ann Davis	turtles	RT	West wing, 1st floor	Emily Clark

Result 173 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
1	15:24:50	SELECT CONCAT_WS(' ', STUDENT_AT.FIRST_NAME_AT, STUDENT_...	3 row(s) returned	0.0073 sec / 0.00002...

2.

```

172 -- For each class level and color show all the student enrolled in alphabetical order per each class (i.e. you start with blue foxes and show all the students)
173 • SELECT CLASS_AT.COLOR_AT AS Color,
174 CLASS_AT.AGE_LEVEL_AT AS Level,
175 CONCAT_WS(' ',STUDENT_AT.FIRST_NAME_AT, STUDENT_AT.MIDDLE_NAME_AT,STUDENT_AT.LAST_NAME_AT ) AS Student_Name,
176 STUDENT_AT.ZIPCODE_AT AS Zipcode
177 FROM STUDENT_AT
178 JOIN CLASS_AT ON STUDENT_AT.CLASS_AT_CODE_AT = CLASS_AT.CODE_AT
179 ORDER BY Color ASC,
180 Level ASC,
181 Student_Name ASC;

```

Color	Level	Student_Name	Zipcode
GF	foxes	Alice Johnson	22030
PM	monkeys	Bob Michael Williams	22046
RT	turtles	Carol Ann Davis	22201

Result 174

	Time	Action	Response	Duration / Fetch Time
1	15:24:50	SELECT CONCAT_WS(' ',STUDENT_AT.FIRST_NAME_AT, STUDENT_...	3 row(s) returned	0.0073 sec / 0.00002...
2	15:28:22	SELECT CLASS_AT.COLOR_AT AS Color, CLASS_AT.AGE_LEVEL_AT...	3 row(s) returned	0.042 sec / 0.00095...

3.

```

182 -- For each parent (caretaker) in alphabetical order (show name and work phone), list all the students enrolled (show name, class level and color)
183 • SELECT CONCAT_WS(' ',PARENT_AT.FIRST_NAME_AT, PARENT_AT.MIDDLE_NAME_AT,PARENT_AT.LAST_NAME_AT) AS Parent,
184 PARENT_AT.WORK_PHONE_AT AS WorkPhone,
185 CONCAT_WS(' ',STUDENT_AT.FIRST_NAME_AT, STUDENT_AT.MIDDLE_NAME_AT, STUDENT_AT.LAST_NAME_AT) AS Student,
186 CLASS_AT.AGE_LEVEL_AT AS Level,
187 CLASS_AT.COLOR_AT AS Color
188 FROM PARENT_AT
189 JOIN RELATIONSHIP_AT ON PARENT_AT.ID_AT = RELATIONSHIP_AT.PARENT_AT_ID_AT
190 JOIN STUDENT_AT ON RELATIONSHIP_AT.STUDENT_AT_ID_AT = STUDENT_AT.ID_AT
191 JOIN CLASS_AT ON STUDENT_AT.CLASS_AT_CODE_AT = CLASS_AT.CODE_AT
192 ORDER BY Parent ASC, Student ASC;

```

Parent	WorkPhone	Student	Level	Color
Jane Johnson	(111) 6667777	Alice Johnson	foxes	GF
Susan Davis	(222) 7778888	Carol Ann Davis	turtles	RT
William JR Williams	(222) 7778888	Bob Michael Williams	monkeys	PM

Result 175

	Time	Action	Response	Duration / Fetch Time
1	15:24:50	SELECT CONCAT_WS(' ',STUDENT_AT.FIRST_NAME_AT, STUDENT_...	3 row(s) returned	0.0073 sec / 0.00002...
2	15:28:22	SELECT CLASS_AT.COLOR_AT AS Color, CLASS_AT.AGE_LEVEL_AT...	3 row(s) returned	0.042 sec / 0.00095...
3	15:29:51	SELECT CONCAT_WS(' ',PARENT_AT.FIRST_NAME_AT, PARENT_AT...	3 row(s) returned	0.049 sec / 0.00037...

4.

```

193 -- For each parent (show name in alphabetical order) list all the teachers (show name in alphabetical order per parent) and their children(students) (show n
194 • SELECT CONCAT_WS(' ',PARENT_AT.FIRST_NAME_AT, PARENT_AT.MIDDLE_NAME_AT, PARENT_AT.LAST_NAME_AT) AS parent,
195 CONCAT_WS(' ',TEACHER_AT.FIRST_NAME_AT, TEACHER_AT.MIDDLE_NAME_AT,TEACHER_AT.LAST_NAME_AT) AS teacher,
196 CONCAT_WS(' ',STUDENT_AT.FIRST_NAME_AT, STUDENT_AT.MIDDLE_NAME_AT, STUDENT_AT.LAST_NAME_AT) AS student
197 FROM PARENT_AT
198 JOIN RELATIONSHIP_AT ON PARENT_AT.ID_AT = RELATIONSHIP_AT.PARENT_AT_ID_AT
199 JOIN STUDENT_AT ON RELATIONSHIP_AT.STUDENT_AT_ID_AT = STUDENT_AT.ID_AT
200 JOIN CLASS_AT ON STUDENT_AT.CLASS_AT_CODE_AT = CLASS_AT.CODE_AT
201 JOIN ASSIGNMENT_AT ON CLASS_AT.CODE_AT = ASSIGNMENT_AT.CLASS_AT_CODE_AT
202 JOIN TEACHER_AT ON ASSIGNMENT_AT.TEACHER_AT_ID_AT = TEACHER_AT.ID_AT
203 ORDER BY parent ASC, teacher ASC, student ASC;

```

parent	teacher	student
Jane Johnson	Katheryn Tiesfamichael	Alice Johnson
Susan Davis	Emily Clark	Carol Ann Davis
William JR Williams	John A Smith	Bob Michael Williams

Result 176

	Time	Action	Response	Duration / Fetch Time
1	15:24:50	SELECT CONCAT_WS(' ',STUDENT_AT.FIRST_NAME_AT, STUDENT_...	3 row(s) returned	0.0073 sec / 0.00002...
2	15:28:22	SELECT CLASS_AT.COLOR_AT AS Color, CLASS_AT.AGE_LEVEL_AT...	3 row(s) returned	0.042 sec / 0.00095...
3	15:29:51	SELECT CONCAT_WS(' ',PARENT_AT.FIRST_NAME_AT, PARENT_AT...	3 row(s) returned	0.049 sec / 0.00037...
4	15:30:59	SELECT CONCAT_WS(' ',PARENT_AT.FIRST_NAME_AT, PARENT_AT...	3 row(s) returned	0.047 sec / 0.00038...